

DIPENDRA SHARMA

Software Development Engineer Intern | Algorithms & Backend

Phone: +91 9780111228 | Email: hello@dipsharma.me | Portfolio: dipsharma.me
[Linkedin](#) | [Github](#) | [Leetcode](#)

Summary

Computer Science student with strong foundations in **Data Structures, Algorithms, Operating Systems, DBMS, Computer Networks, and OOP**. LeetCode **peak rating 1570, 313+ problems solved, 500-Day Badge**. Confident in Arrays, Strings, Hash Tables, Two Pointers, Sliding Window, Binary Search, Linked Lists, Stacks, Queues, and Greedy. I believe a problem well-stated is half-solved — I focus on understanding depth before writing code, and always aim for the most optimal solution first. I love building cool things that work in the real world — currently focused on backend systems, database internals, and Linux. Following NeetCode 150 and studying foundational books by **Boris Cherny (TypeScript)** and industry-standard documentation.

Competitive Programming & Problem Solving

LeetCode: Peak Rating **1570** | **313+ problems solved** | **500-Day Badge** | 100-Day Badge 2026 | 3+ contests
Strong Topics: Arrays, Strings, Hash Tables, Two Pointers, Sliding Window, Binary Search, Linked Lists, Stacks, Queues, Greedy
Patterns Mastered: Kadane's, Two-Pointer, Sliding Window, Binary Search on Answer, Dutch National Flag, Prefix Sum, Hash Map Counting, Monotonic Stack, Fast/Slow Pointers
Approach: Analyze time/space complexity first | target $O(N)$ or $O(\log N)$ after brute force | iterate to optimal
Practice: Daily LeetCode streak | NeetCode 150 | university coding competitions

Technical Skills

Languages: C++, Java, JavaScript, TypeScript, SQL
Core CS: **OS** (Processes, Threads, Scheduling, Memory), **DBMS** (Normalization, ACID, Indexing, Transactions), **CN** (TCP/UDP, HTTP, DNS), **OOP** (Polymorphism, Inheritance, Encapsulation)
Linux: Kernel fundamentals (process, memory, file systems, permissions), systemd, cron, SSH, bash scripting, log analysis (journalctl, tail), networking tools (ss, netstat), package management, file operations (grep, awk, sed, find, xargs)
Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, PostgreSQL, Prisma, MongoDB
Frontend: React, Next.js, TypeScript, Redux, Tailwind CSS
DevOps: Docker, Git, GitHub Actions, AWS (EC2, S3, IAM), Vercel, Render

Projects

Yeti Jobs | [GitHub](#) | [Live](#)

Jan 2026 – Apr 2026

- Build **50+ REST APIs** with RBAC — hash-map based role validation for $O(1)$ permission checks across admin, recruiter, and candidate workflows.
 - Optimized PostgreSQL via composite indexing (tree-structured search), cutting query time **30%** (7ms → 5ms per EXPLAIN ANALYZE); connection pooling handled **20 concurrent DB connections**.
 - Load-tested with Apache Bench: **1,000 requests** at **100 concurrent users** → **67 req/sec**, p95 **1,929ms**, **0 failures** on a free-tier Render instance.
 - Multi-stage Docker builds reduced image size by **76%** (1.99GB → 520MB backend, 1.03GB → 220MB frontend); cron job eliminated cold starts (**50s** → **<2s**), sustaining **99.5% uptime** over 3 months.
 - 28+ integration tests (Jest + Supertest); deployed via **Vercel + Render** with **GitHub Actions CI/CD**.
- Tech:** PostgreSQL, Node.js, Express, React, TypeScript, Docker, JWT, Jest, GitHub Actions

Achievements

- **CodeaThon 7.0 Winner** — solved 4/5 algorithm questions, 1st place (cash prize, 120+ participants).
- LeetCode peak rating **1570** — 313 problems solved, 500-Day Badge, 100-Day Badge 2026.
- Selected Contributor — **GirlScript Summer of Code 2026** (open-source).
- Led multiple hackathon teams of 3-6 (SIH, Digimantra, Nepal Police) — shipped prototypes under 24-hour deadlines.
- **GitHub:** 2000+ commits (last 12 months) | **5+ merged PRs** including **Zed editor** contributions.
- **CEED Member** — Entrepreneurship Program, Chitkara University.

Certifications

- **AWS Educate Introduction to Cloud 101** — EC2, S3, IAM fundamentals
- **C++ Code Like You Are in MATRIX** — Memory Management, Data Structures, Algorithms

Education

Chitkara University

Aug 2024 – Expected 2027

Bachelor of Computer Applications (BCA)

CGPA: 9.38/10

– Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, Software Engineering